Fourier transform

1. Complex number. $R + jI$.

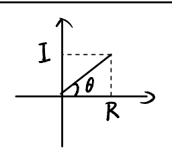
↙ reparametrization

$r(\cos\theta + j\sin\theta)$

where $r = \sqrt{I^2 + R^2}$.

$\theta = \tan^{-1}(\frac{I}{R})$

$= re^{j\theta}$ (euler's formula)



Conjugate. $C^* = R - jI$

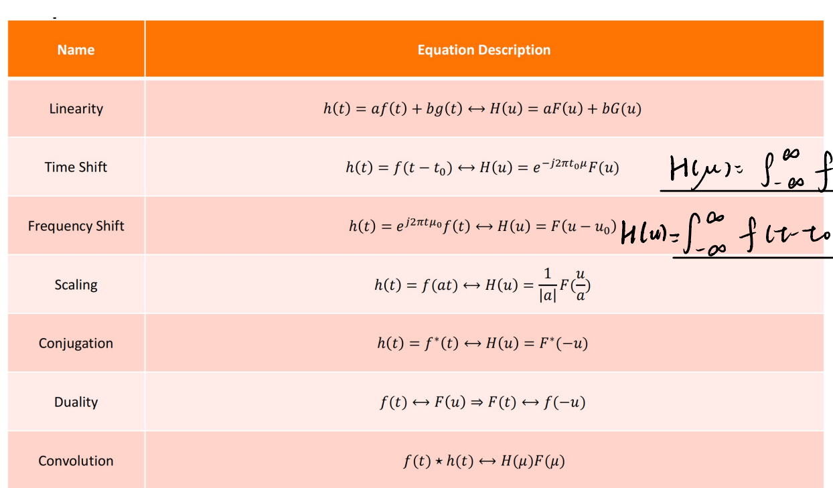
Fourier Series

$f(t)$: periodic function with period T.

$$f(t) = \sum_{n=-\infty}^{n=+\infty} C_n e^{j\frac{2\pi}{T} n t}.$$

$$= \sum_{n=-\infty}^{n=\infty} C_n \left(\cos(\frac{2\pi n}{T})t + j\sin(\frac{2\pi n}{T})t \right).$$

where $C_n = \frac{1}{T}\int_{-T/2}^{T/2} f(t) e^{-j\frac{2\pi n}{T} t} dt.$



Name	Equation Description
Linearity	$h(t) = af(t) + bg(t) \leftrightarrow H(u) = aF(u) + bG(u)$
Time Shift	$h(t) = f(t - t_0) \leftrightarrow H(u) = e^{-j2\pi t_0 \mu} F(u)$
Frequency Shift	$h(t) = e^{j2\pi t \mu_0} f(t) \leftrightarrow H(u) = F(u - u_0)$
Scaling	$h(t) = f(at) \leftrightarrow H(u) = \frac{1}{\lvert a\rvert} F(\frac{u}{a})$
Conjugation	$h(t) = f^*(t) \leftrightarrow H(u) = F^*(-u)$
Duality	$f(t) \leftrightarrow F(u) \Rightarrow F(t) \leftrightarrow f(-u)$
Convolution	$f(t) \star h(t) \leftrightarrow H(\mu)F(\mu)$

$H(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt.$

$H(u) = \int_{-\infty}^{\infty} f(t-t_0) e^{-j2\pi u t} dt = e^{-j2\pi u t_0} \int_{-\infty}^{\infty} f(t-t_0) e^{-j2\pi u(t-t_0)} dt$

$= e^{j2\pi u t_0} F(u)$

Fourier Transform

$$\begin{cases} F(\mu) = \mathcal{J}\{f(t)\} = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt. \\ f(t) = \mathcal{J}^{-1}\{F(\mu)\} = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu. \end{cases}$$

e.x. $f(t) = \begin{cases} A, & -W/2 \le t \le W/2. \\ 0. & \text{otherwise.} \end{cases}$

$$F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt.$$

$$= \int_{-W/2}^{W/2} A e^{-j2\pi\mu t} dt.$$

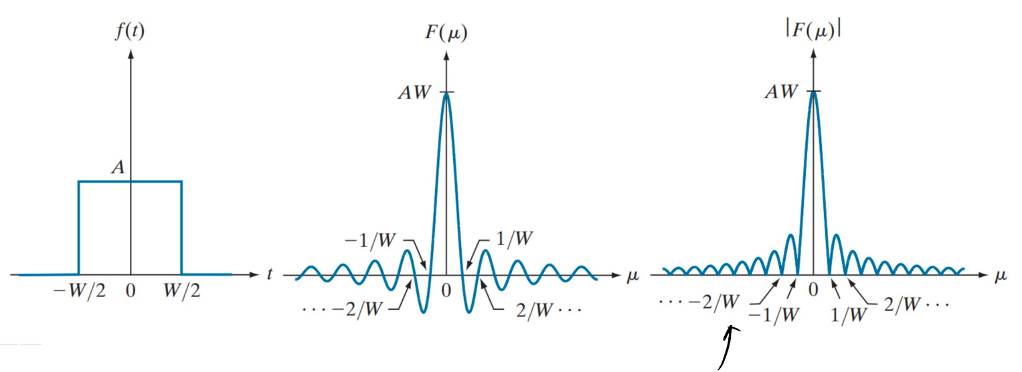
$$= \frac{A}{-2j\pi\mu} e^{-j2\pi\mu t} \Big|_{-W/2}^{W/2}$$

$$= \frac{A}{-2j\pi\mu} [e^{-j\pi\mu W} - e^{j\pi\mu W}]$$

$$= \frac{A}{2j\pi\mu} [\cos(\pi\mu W) + j\sin(\pi\mu W) - \frac{1}{\cos(\pi\mu W) + j\sin(\pi\mu W)}]$$

$$= \frac{A}{2j\pi\mu} [\cos(\pi\mu W) + j\sin(\pi\mu W) - \cos(\pi\mu W) + j\sin(\pi\mu W)]$$

$$= \frac{A}{2j\pi\mu} \cdot 2j\sin(\pi\mu W) = \frac{A}{\pi\mu}\sin(\pi\mu W) = AW\frac{\sin(\pi\mu W)}{\pi\mu W}$$



↑ "spectrum"

Convolution: $f(t) * h(t) = \int_{-\infty}^{\infty} f(\tau) h(t-\tau) d\tau.$

↓ Fourier Transform.

$$\mathcal{J}\{f(t) * h(t)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\tau) h(t-\tau) d\tau \, e^{-j2\pi\mu t} dt.$$

$$= \int_{-\infty}^{\infty} f(\tau) \left[\int_{-\infty}^{\infty} h(t-\tau) e^{-j2\pi\mu t} dt\right] d\tau.$$

$\mathcal{J}\{h(t-t_0)\} = e^{-j2\pi t_0 \mu} F(\mu).$

$$= \int_{-\infty}^{\infty} f(\tau) [H(\mu) e^{-j2\pi\mu\tau}] d\tau.$$

$$= H(\mu) \int_{-\infty}^{\infty} f(\tau) e^{-j2\pi\mu\tau} d\tau.$$

$$= H(\mu) \cdot F(\mu)$$

$$\Rightarrow \begin{cases} f(t) * h(t) \Leftrightarrow H(\mu) \cdot F(\mu) \\ f(t) h(t) \Leftrightarrow H(\mu) * F(\mu) \end{cases}$$

Discrete Fourier Transform (DFT)

spatial → frequency

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j\frac{2\pi u x}{M}}, \quad u = 0, 1, \dots, M-1.$$

frequency → spatial

$$f(x) = \frac{1}{M}\sum_{x=0}^{M-1} F(u) e^{j\frac{2\pi u x}{M}}. \quad x = 0, 1, \dots, M-1$$

$$= \underbrace{\frac{1}{M}\sum_{x=0}^{M-1} F(u)}_{\text{Scaling para.}} \underbrace{\{\cos(\frac{2\pi u}{M}x) + j\sin(\frac{2\pi u}{M}x)\}}_{\text{wave component}}$$

2D. DFT

$$F(u,v) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}.$$

$$f(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} F(x,y) e^{j2\pi(\frac{ux}{M} + \frac{vy}{N})}.$$

remark:

$F(0,0) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y) = MN\bar{f}$

$$\begin{cases} F(u+k_1M, v+k_2N) = F(u,v) \\ f(u+k_1M, v+k_2N) = f(u,v) \end{cases} \text{"periodic"}$$

$$\begin{cases} f(x,y) e^{j2\pi(\frac{u_0x}{M} + \frac{v_0y}{N})} = F(u-u_0, v-v_0) \\ F(u,v) e^{-j2\pi(\frac{ux_0}{M} + \frac{vy_0}{N})} = f(x-x_0, y-y_0) \end{cases}$$

$$F(r, \theta) = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(r,\theta) e^{-jwr\cos(\theta-\phi)}$$

$$\mathcal{J}\{f(r, \theta+\theta_0)\} = \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(r, \theta+\theta_0) e^{-jwr\cos(\theta-\phi)}$$

$$= \sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(r, \alpha) e^{-jwr\cos(\alpha-(\phi+\theta_0))}$$

$$= F(w, \phi+\theta_0)$$

rotate the image ⇔ rotate the fouriour Spectrum

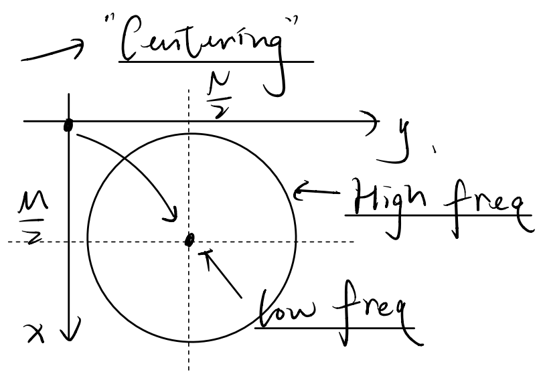
Centering DFT

1D: $f(x) e^{j2\pi(\frac{u_0x}{M})} \Leftrightarrow f(u-u_0)$

$\Rightarrow f(x) e^{j2\pi(\frac{\frac{M}{2}x}{M})} = f(x) e^{j\pi x} = f(x)(-1)^x \Leftrightarrow f(u - \frac{M}{2})$

2D: $f(x,y) e^{j2\pi(\frac{u_0x}{M} + \frac{v_0y}{N})} \Leftrightarrow F(u-u_0, v-v_0)$

$\Rightarrow f(x,y)(-1)^{x+y} \Leftrightarrow F(u - \frac{M}{2}, v - \frac{N}{2})$ → "Centering"



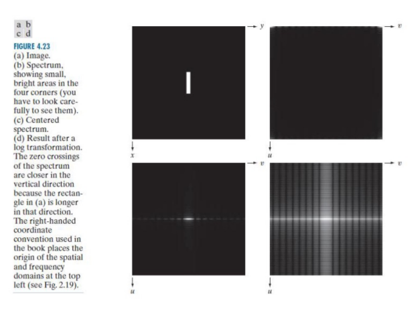
Fouriour Spectrum & Phase Angle

$$F(u,v) = R(u,v) + jI(u,v)$$

$$= |F(u,v)| e^{j\phi(u,v)}$$

where:

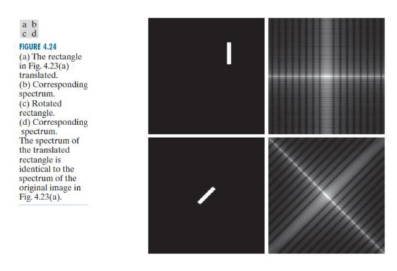
$$\begin{cases} |F(u,v)| = \sqrt{R^2(u,v) + I^2(u,v)} \Rightarrow \text{Fourier Spectrum} \\ \phi(u,v) = \tan^{-1}(\frac{I(u,v)}{R(u,v)}) \Rightarrow \text{Phase Angle} \end{cases}$$



a b
c d

FIGURE 4.23 (a) Image. (b) Spectrum, showing small, bright areas in the four corners (you have to look carefully to see them). (c) Centered spectrum. (d) Result after a log transformation. The zero crossings of the spectrum are closer in the vertical direction because the rectangle in (a) is longer in that direction. The right-handed coordinate convention used in the book places the origin of the spatial and frequency domains at the top left (see Fig. 2.19).

亮纹越宽 ⇔ 频谱亮纹越窄



a b
c d

FIGURE 4.24 (a) The rectangle in Fig. 4.23(a) translated. (b) Corresponding spectrum. (c) Rotated rectangle. (d) Corresponding spectrum. The spectrum of the translated rectangle is identical to the spectrum of the original image in Fig. 4.23(a).

亮纹平移 → 频谱不变

亮纹旋转 → 频谱同向旋转

$$f(x,y) = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v) e^{(\frac{ux}{M} + \frac{vy}{N}) \cdot j2\pi}$$

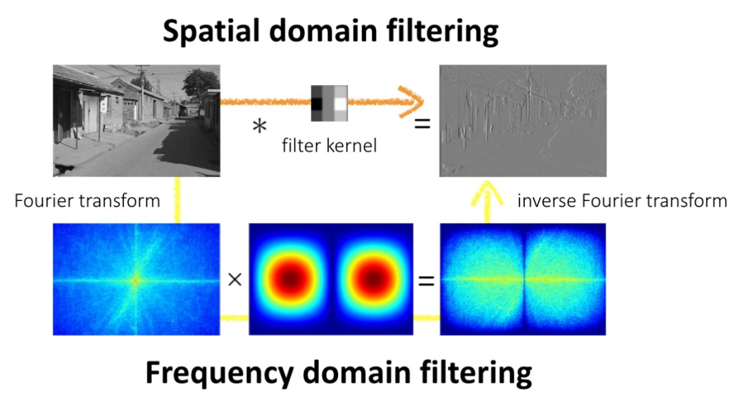
$$= \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} |F(u,v)| e^{2\pi j(\frac{ux}{M} + \frac{vy}{N} + \frac{\phi(u,v)}{2\pi})}$$

↙ Amplitude of sinusoid with freq (u,v)

↓ Displacement with respect to origin

More information

Filtering in Frequency Domain



$(f * h)(x,y) = F(u,v) H(u,v)$

Convolution in Spatial ⇔ Mult in Frequential

$(f \cdot h)(x,y) \Leftrightarrow \frac{1}{MN}(H * F)(u,v)$

Mult in Spatial ⇔ Convolution in Freq

Basic Filtering

$$f(x,y) \xrightarrow{DFT} F(u,v) \xrightarrow{*H(u,v)} F(u,v) \cdot \underbrace{H(u,v)}_{\text{Filter}} \xrightarrow{(DFT)^{-1}} g(x,y).$$

① Smoothing Filter (Lowpass).

1. ideal lowpass filter

$$H(u,v) = \begin{cases} 1, & \text{if } D(u,v) \le D_0. \\ 0, & \text{if } D(u,v) > D_0 \end{cases}$$

$D(u,v) = \sqrt{(u - \frac{P}{2})^2 + (v - \frac{Q}{2})^2}$ where image: $P \times Q$

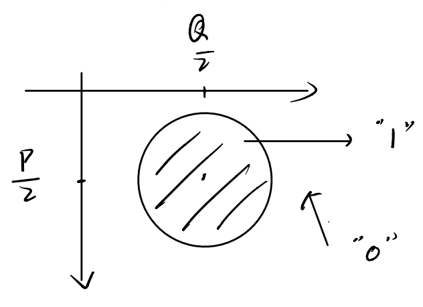


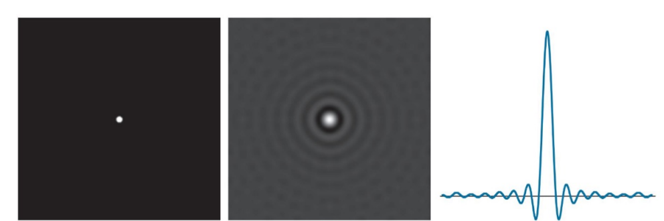
Image Power: $P_T = \sum_u\sum_v P(u,v)$

$P(u,v) = I^2(u,v) + R^2(u,v)$

percent of power

$$\alpha = \sum_u\sum_v \frac{P(u,v)}{P_T} \times 100\%.$$

D_0 ↓ more blurred



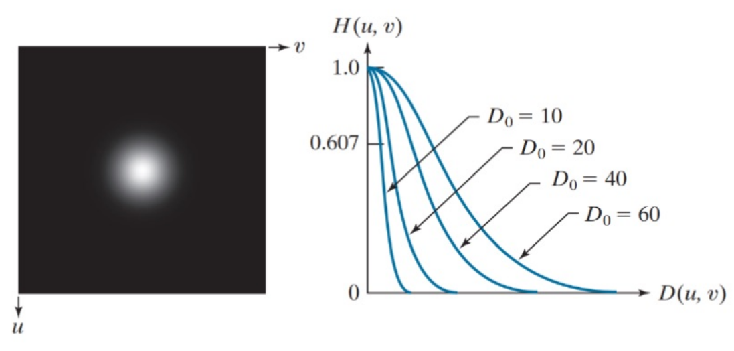
ILPF ⇒ sharp cutoff in freq ⇒ Ripple in Spatial ⇒ ringing artifact

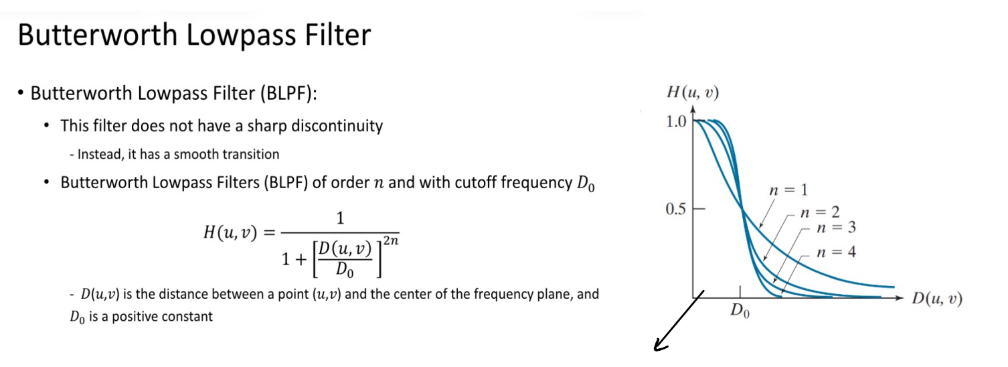
↓

What we need: smooth cutoff

Gaussian Lowpass Filter

$H(u,v) = e^{-D^2(u,v)/2\sigma^2} = e^{-D^2(u,v)/2D_0^2}.$





Butterworth Lowpass Filter

- Butterworth Lowpass Filter (BLPF):
  - This filter does not have a sharp discontinuity
    - Instead, it has a smooth transition
  - Butterworth Lowpass Filters (BLPF) of order n and with cutoff frequency D_0

$$H(u,v) = \frac{1}{1 + \left[\frac{D(u,v)}{D_0}\right]^{2n}}$$

- $D(u,v)$ is the distance between a point (u,v) and the center of the frequency plane, and D_0 is a positive constant

n ↓ more Smooth. ringing ↓

n ↑ preserve more low freq ringing ↑.

Sharpening Filter (Highpass).

$H_{hp}(u,v) = 1 - H_{lp}(u,v)$

Ideal Highpass Filter

$$H(u,v) = \begin{cases} 0, & D(u,v) \le D_0. \\ 1 & D(u,v) > D_0 \end{cases}$$

- A 2-D Gaussian highpass filter (GHPL) is defined as:

$$H(u,v) = 1 - e^{-\frac{D^2(u,v)}{2D_0^2}}$$

- A 2-D Butterworth highpass filter (BHPL) is defined as:

$$H(u,v) = \frac{1}{1 + \left[\frac{D_0}{D(u,v)}\right]^{2n}} \longleftrightarrow \text{BLPF}: H(u,v) = \frac{1}{1 + (\frac{D(u,v)}{D_0})^{2n}}$$

Unsharp Masking, Highboost Filtering. High Frequency-Emphasis

$g_{mask}(x,y) = f(x,y) - f_{LP}(x,y)$

$f_{LP}(x,y) = \mathcal{T}^{-1}\{F(u,v) H_{LP}(u,v)\}$

$g(x,y) = f(x,y) + k g_{mask}(x,y)$

$\Rightarrow g(x,y) = \mathcal{T}^{-1}\{F(u,v) + k[F(u,v) - F(u,v) H_{LP}(u,v)]\}.$

$= \mathcal{T}^{-1}\{F(u,v)(1 + k(1 - H_{LP}(u,v)))\}$

✓ More general.

$g(x,y) = \mathcal{T}^{-1}\{(k_1 + k_2 H_{HP}(u,v)) F(u,v)\}.$

k_2 ↖ contribution of high freq

k_1 ↑ contribution of origin image